

Intro to Beams

1 Background Knowledge Check

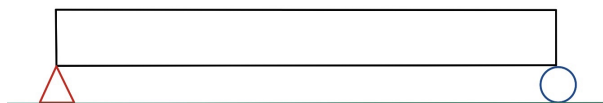
Must be familiar with Newton's 3 laws, forces, vectors and moments/torques.

2 Beams in physics

The beams discussed in this document will all be assumed to be static (in static equilibrium/not moving). We will be taking a look on the different types of forces you may see acting upon beams and setting up the equations to calculate missing forces and/or lengths.

3 Newton's 3rd Law and Supports

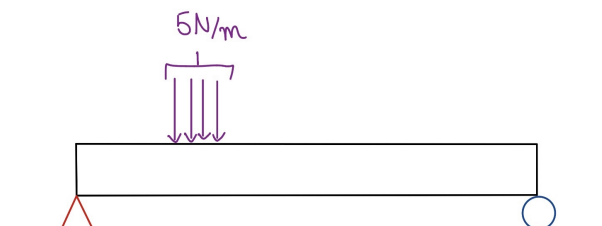
Newton's 3rd Law is essential when working with beams. Often, you may see images that look like this:



In these diagrams, a triangular support is known as a pin or hinge support. It provides both a horizontal and vertical reaction force. Whereas the circular support, known as a roller support provides only a vertical reaction force. These forces are the reaction forces that these supports apply to the beam because of the beam's weight pushing upon them. There is also a third type of support, a fixed end support, which exerts both vertical and horizontal reaction forces like the pin support as well as a moment on the beam.

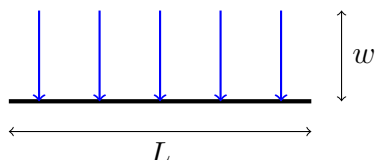
4 Distributed Loads

You may see multiple forces working over a distance on the beam such as follows:



We call these distributed loads. Rather than dealing with all these forces by themselves (which would make the equation super long), we instead replace these loads with one singular force. We call this force an equivalent force because if that force actually was acting on the beam, the effects of it on the beam would be the same as that of the load on the beam. The equations for each different kind of loads, their equivalent forces and locations are as follows where w is the force per unit length, L is the length over which the force acts and w_{max} (used for Triangular loads) is the maximum force in the load:

Uniformly Distributed Load/UDL

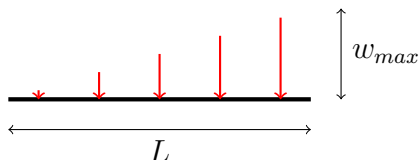


Rectangular Load

Force: $F_{eq} = w \cdot L$

Location: $\bar{x} = \frac{L}{2}$ (Center)

Linearly Varying Load/LVL



Triangular Load

Force: $F_{eq} = \frac{1}{2}w_{max}L$

Location: $\bar{x} = \frac{1}{3}L$ (from heavy end)

5 Static Equilibrium Equations

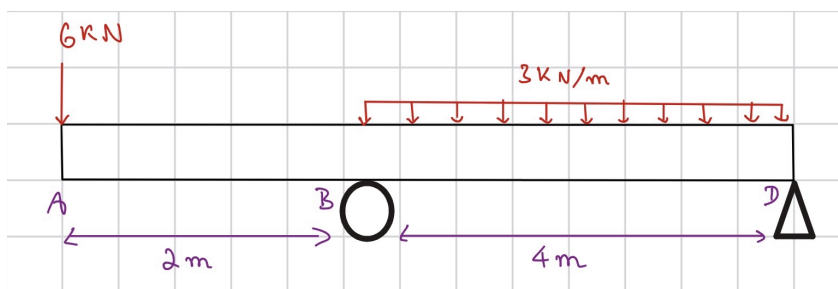
Once all distributed loads have been converted into their equivalent forces, you can solve for the unknown reaction forces at the supports using the three conditions of equilibrium:

1. $\sum F_x = 0$ (Sum of all horizontal forces must be zero)
2. $\sum F_y = 0$ (Sum of all vertical forces must be zero)
3. $\sum M_A = 0$ (Sum of all moments about any point A must be zero)

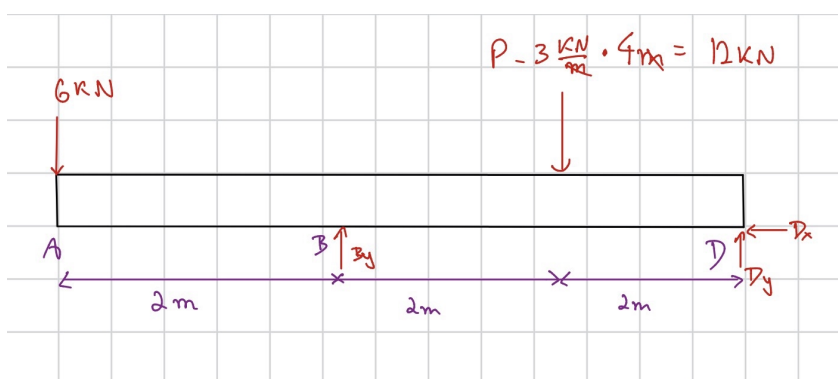
We would then extend these equations by adding all of our forces in the x direction, forces in the y direction and moments. When doing so, we may assume a direction for which we think our unknown force or moment is acting on the beam. Sometimes, our assumptions about the directions we choose may be wrong. We will know our assumption was wrong if we get a negative answer for the magnitude of our calculated forces or moment. In such cases, we simply write "assumption was wrong" and write a statement about the correct direction of our force or moment

Let's try and walk through the following problem:

1. Find the reaction forces at B and D for the following beam:

**Solution:**

- When encountering these problems, we always start by drawing a free body diagram. For this we include our object of focus (the beam) and any forces acting upon it as well as any relevant length measurements. Moreover, we will also replace the UDL with an equivalent point force:



- Next, we start to write our three static equilibrium equations; again, these will all be equal to 0. We know that because our beam isn't moving:

- The only horizontal force acting upon the beam is D_x . Therefore, the magnitude of D_x must be 0:

$$\sum F_x = 0 = D_x$$

- We are currently assuming that both of our vertical reaction forces point upwards. This would make sense because these forces must be reacting to the 6 and 12 kN forces acting downward upon the beam. Choosing up as positive, we get:

$$\sum F_y = 0 = -6 \text{ kN} - 12 \text{ kN} + B_y + D_y$$

$$B_y + D_y = 6 \text{ kN} + 12 \text{ kN}$$

We have two unknowns but only one equation, so we can't solve just yet

- We now pick either of the spots where our reaction forces act (B or D) to make the moment equation. We choose these spots because this allows the force that acts upon that point to not appear in our equation. So we can solve for the other missing force (Note m stands for meters):

$$\sum M_B = 0 = (6 \text{ kN} \cdot 2 \text{ m}) - (12 \text{ kN} \cdot 2 \text{ m}) + (D_y \cdot 4 \text{ m})$$

$$0 = 12 \text{ kN} \cdot \text{m} - 24 \text{ kN} \cdot \text{m} + 4D_y \text{m}$$

$$\cancel{\sum}(4D_y) \cancel{\text{m}} = \cancel{\sum} 12 \text{ kN} \cdot \cancel{\text{m}}$$

$$D_y = 3 \text{ kN}$$

Therefore, we found that D_y is 3 kN and our assumption was correct (positive solution), so it points up!

Now we have to find B_y . For this, we are going to use our $\sum F_y$ equation again and sub in our value for D_y :

$$B_y + 3 \text{ kN} = 6 \text{ kN} + 12 \text{ kN}$$

$$B_y = 6 \text{ kN} + 12 \text{ kN} - 3 \text{ kN}$$

$$B_y = 15 \text{ kN}$$

Therefore, we found all of our missing forces. Sometimes, professors may ask you to confirm your findings using a fourth equation. To do this you simply set up another moment equation, but this time about a different point (For example D) and confirm that adding all the moments does indeed give you 0.